

FORCED CONVECTION DRYING

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Objective:

The aim of this experiment is to analyse the drying process and different stages of drying in a forced convection drying.

Theory and Background:

Drying is an important industrial operation. Drying is generally performed to remove the impurities from the product obtained from processes. Drying is used to separate the solvent from the solid/semi-solid products. In the process of drying, the solvent involved is water (mostly). Drying is done by evaporation of the solvent into another agent (mostly dry air). Since evaporation requires heat, the air supplied is heated.

Forced convection drying is a method for drying materials more rapidly through the application of forced hot air directed by fans or blowers. Unlike natural drying, where air moves passively, forced convection speeds up the drying process, making it more efficient and uniform. The rate of drying is affected by the moisture present, the nature of the material, the bonding of solid molecules with the moisture, the area of the surface exposed, and the thermal conductivity of the solid.

In this experiment, we are using air drying, where heated air removes moisture from various food items. The hot, dry air is blown over the moist food, causing water to evaporate into the air. The airflow then carries the moisture away. Since the air used is completely dry, a significant chemical potential gradient is maintained between the food and the air, enhancing the drying process. As the moisture content decreases, the drying rate changes. In this experiment, we measure the drying rate and moisture content at different time intervals to analyze how drying progresses over time.

Drying process have 3 stages:

1. **Constant rate stage:** When we start the drying process, at the beginning the rate of removal of moisture remains steady. This is because of the surface moisture. The surface moisture evaporates quickly and efficiently. During this time, the temperature of the material and the air temperature remains relatively unchanged.
2. **Falling rate stage:** As the surface moisture decreases, the drying rate decreases as the moisture from the layer deeper has to come to the surface for evaporation. The movement of moisture from deeper to surface layer is slow due to the resistance provided by solid particles.
3. **Equilibrium stage:** When most of the moisture is gone, the drying rate drops to nearly zero. At this point, evaporation is minimal, indicating that the material is almost completely dry.

Apparatus:

The forced convective drying apparatus consists of:

1. Digital thermometers
2. Heater
3. Blower
4. Digital ammeter
5. Digital voltmeter
6. Tray
7. Stand
8. Around 20 gm of dry sprouted beans.



Figure 1: Schematic of the apparatus

Procedure:

1. Take dry sprouted beans (about 20 grams).
2. Weigh the dry sprouted beans.

3. Soak dry sprouted beans in water for over a day so that they absorb a certain amount of moisture.
4. Take beans out of the water.
5. Measure the weight of the wet beans to get the initial amount of moisture inside these beans.
6. Put the beans in a tray inside the dryer and start the blowing air and heater so that hot air blows over the beans.
7. Weigh the beans after various periods of time so that the drying rate and moisture content may be estimated for the beans.

Known parameters:

- *Mass of the dry beans (S) : 20 g*
- *Mass of the tray : 110 g*
- *Area of the tray : 58.0644 cm²*
- *Mass of tray with soaked beans : 160 g*

Formulae:

- *Moisture content (X) = $\frac{W-S}{W}$*
 - *Where, W is the weight of beans (soaked)*
 - *S is the weight of dry beans*
- *Drying rate : $N_A = - \left(\frac{S}{A} \right) * \left(\frac{-dX}{dt} \right)$*
 - *Where S is the weight of dry beans*
 - *A is the area of the try*
 - *X is the moisture content*
 - *t is the time*

Readings:

TABLE 1: TABLE CONTAINING EXPERIMENTAL DATA

Time (min)	Weight of tray + beans (g)
0	160
10	159.9
20	159.6
30	159.3
40	159

50	158.7
60	158.4
70	158.2
80	158
90	157.8
100	157.6

TABLE 2: TABLE CONTAINING CALCULATED DATA

Time t (min)	Weight of tray + beans (g)	W (Weight of beans) (g)	Moisture content (X) = $\frac{W-S}{W}$	$\left(\frac{-dX}{dt}\right) \left(\frac{1}{s}\right)$	Drying rate $N_A - \left(\frac{S}{A}\right) * \left(\frac{-dX}{dt}\right)$ $\left(\frac{g}{cm^2 s}\right)$
0	160	50	0.6		
10	159.9	49.9	0.5991983968	0.000001336005344	0.00001069
20	159.6	49.6	0.5967741935	0.000004040338742	0.00003232
30	159.3	49.3	0.5943204868	0.000004089511222	0.00003272
40	159	49	0.5918367347	0.000004139586869	0.00003312
50	158.7	48.7	0.5893223819	0.000004190587939	0.00003352
60	158.4	48.4	0.5867768595	0.000004242537377	0.00003394
70	158.2	48.2	0.5850622407	0.000002857698067	0.00002286
80	158	48	0.5833333333	0.000002881512218	0.00002305
90	157.8	47.8	0.5815899582	0.000002905625291	0.00002325
100	157.6	47.6	0.5798319328	0.00000293004231	0.00002344

Results and discussion:

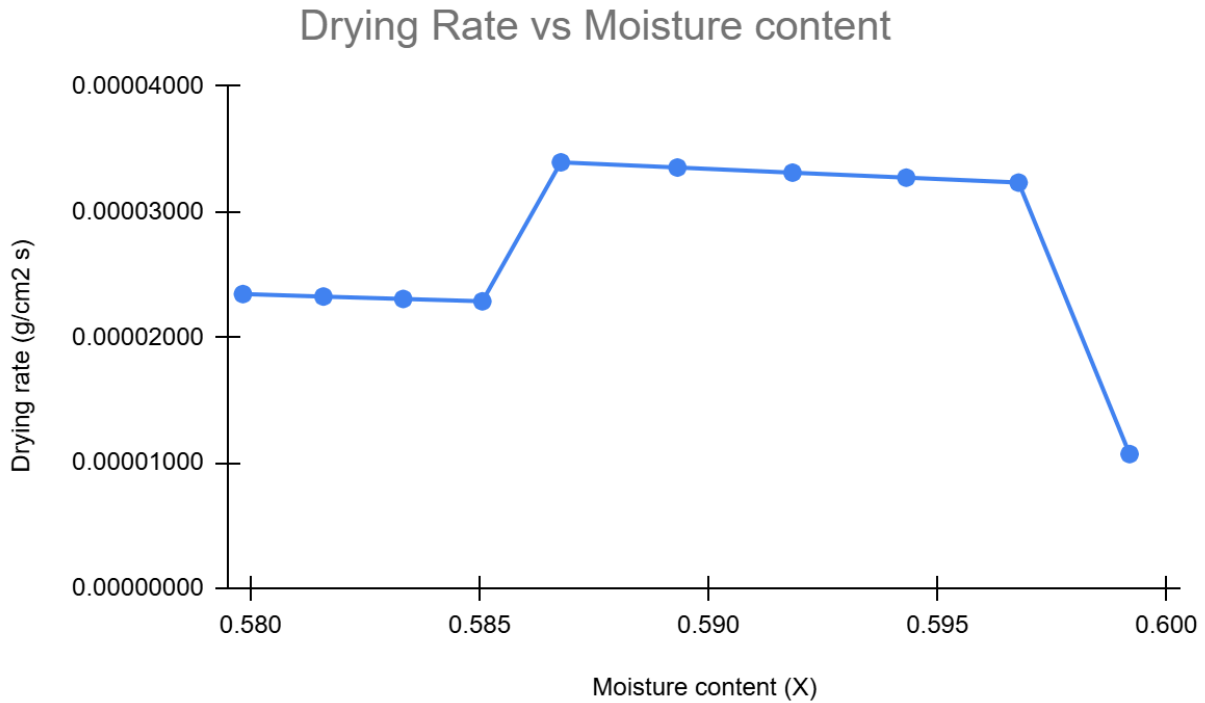


Figure 2: Figure showing the drying rate vs moisture content

From figure 2 we can see that the rate of drying is not constant; it increased, remained constant for sometime and then decreased. The trends seen are:

1. The initial moisture content was 0.6 and gradually declined to 0.5798 after 100 minutes.
2. During the first 60 minutes, the drying rate remained relatively high, peaking at $0.00003394 \frac{g}{cm^2 s}$. This suggests the constant rate period, where surface moisture evaporates steadily.
3. After 60 minutes, the drying rate significantly dropped to $0.00002286 \frac{g}{cm^2 s}$ and continued decreasing, indicating the falling rate period, where moisture diffusion from within the beans slows the process.

The graph in Figure 2 follows the expected drying pattern. At first, the drying rate stays relatively stable, confirming the constant rate period, where surface moisture evaporates efficiently. As moisture content decreases, the drying rate suddenly decreases around $X = 0.585$, marking a transition where internal moisture diffusion takes over.

After this peak, the drying rate declines gradually, indicating the falling rate period. However, instead of dropping continuously, it stabilizes again, likely due to the system reaching a near-equilibrium state. Theoretically, the drying rate should keep decreasing until all moisture is

removed, but in real conditions, the air isn't completely dry. Some residual moisture in the ambient air could have slowed further evaporation, causing the drying rate to remain constant.

Conclusion:

The drying experiment clearly demonstrated the different phases of the drying process. Initially, during the constant rate period, surface moisture evaporated efficiently, maintaining a steady drying rate. As drying progressed, the falling rate period began, where moisture migration from within the material slowed down the process. The drying rate peaked around $X = 0.585$, marking a transition where internal moisture diffusion became dominant. After this peak, the drying rate declined, eventually stabilizing in the equilibrium stage, where moisture removal became minimal. The results align with standard drying theory, confirming the shift from surface-controlled evaporation to diffusion-controlled drying, with external conditions influencing the final equilibrium state.

Sources of errors:

- The weighing scale may not be having more significant digits to measure the exact weight.
- There could have been fluctuations in the ambient temperature and humidity.

References:

- [1] “Convective Drying - an overview | ScienceDirect Topics,” www.sciencedirect.com.
<https://www.sciencedirect.com/topics/engineering/convective-drying>
- [2] ‘Lab Manual’ provided in the classroom.
- [3] A. Belghit, M. Kouhila, and B. C. Boutaleb, “Experimental study of drying kinetics by forced convection of aromatic plants,” *Energy Conversion and Management*, vol. 41, no. 12, pp. 1303–1321, Mar. 2000, doi: [https://doi.org/10.1016/S0196-8904\(99\)00162-4](https://doi.org/10.1016/S0196-8904(99)00162-4).
- [4] R. Treybal, “Mass-transfer operations,” McGraw-Hill, New York, 3rd ed., pp. 275–313, 1980.
- [5] A. Tesfamicha and A. Assefa, “Experimental Analysis of Potato Slices Drying Characteristics using Solar Dryer,” *Journal of Applied Sciences*, vol. 13, no. 6, pp. 939–943, Jun. 2013, doi: <https://doi.org/10.3923/jas.2013.939.943>.